

GeoForge Desktop User Manual

Make scientific models *actually run.*

From a scientific question to a reproducible model run. This book follows the interface you actually use and explains what happens, what the agent handles, and which decisions still belong to you.

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00 Three ideas to remember



Chat is primary

Describe the scientific task in normal language. GeoForge turns it into a visible, reviewable workflow.



A KI is executable guidance

A KI tells the agent how to install, prepare, run, check, and interpret one scientific model.

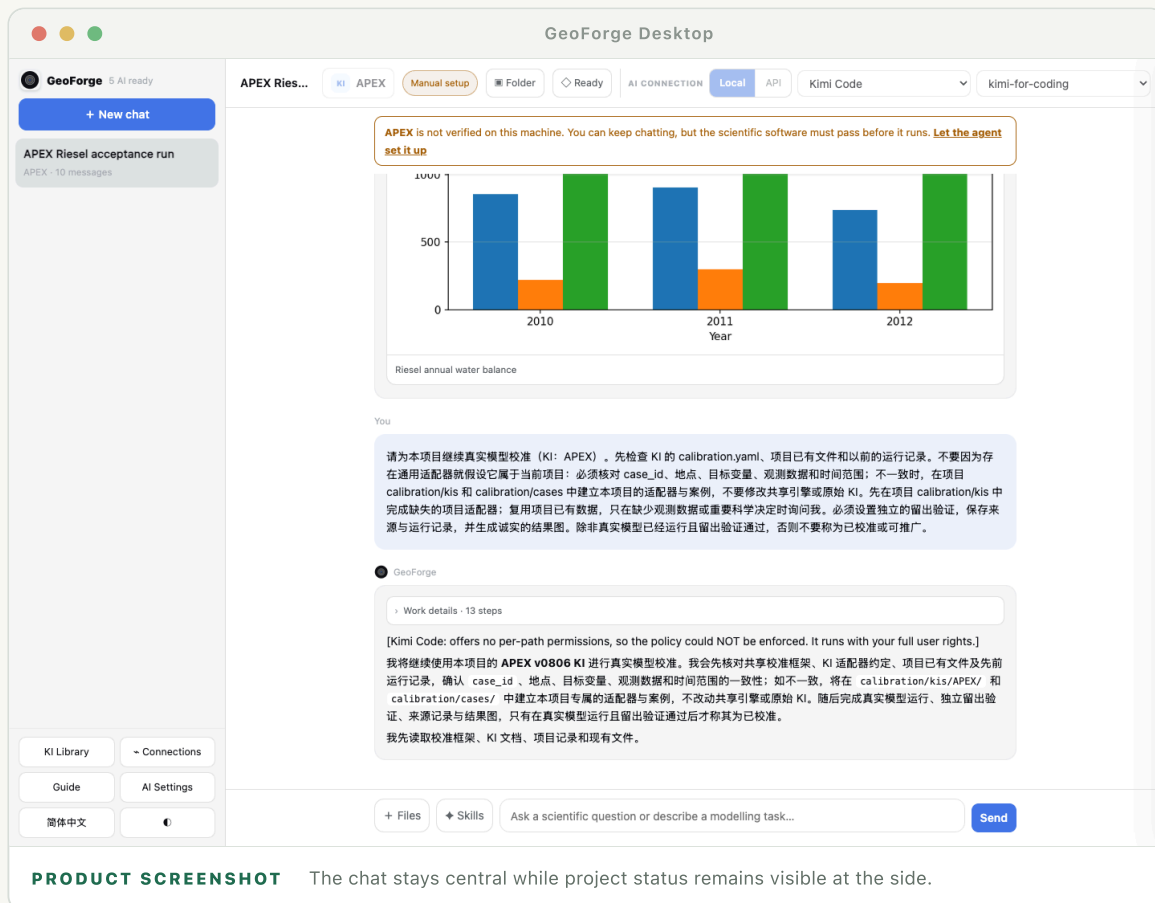


Every project stays local

Messages, source data, model inputs, outputs, plots, and provenance live together in one folder on your Mac.

Your first successful run

The shortest route is: create a local project, describe a concrete scenario, let the agent prepare what it safely can, answer only the decisions that truly need you, and verify the saved result.



Five actions

- 1 Choose New conversation. Confirm the suggested project folder or select another writable folder.
- 2 Write one concrete request: place, period, process, and desired result. Missing details can be decided later.
- 3 Confirm the KI proposed by the agent. A KI may be available before its scientific software is verified on this Mac.
- 4 Open Project data. Use Resolve with agent for public or derivable data; add private files yourself when needed.
- 5 Let the agent run the real model. Review plots and metrics, then open the project folder to inspect the actual files.

How to know the run is real

- ✓ The project shows a completed model-run stage, not only a written plan.
- ✓ The outputs folder contains files created by the scientific model.
- ✓ The result cites the executable, input files, dates, parameters, and validation checks used.
- ✓ Plots are linked to saved data rather than being unsupported illustrations.

TIP

A polished answer is not proof of a run. Saved inputs, command provenance, outputs, and checks are the proof.

Worked example: run APEX for a 50 ha rainfed corn field

This chapter is the practical spine of the manual. Follow one real acceptance case near Riesel, Texas, for 2010–2012. You will create a local project, choose the APEX KI, let the agent prepare public weather and model inputs, respond to one data decision, run the verified APEX0806 executable, inspect the plots, and confirm the saved files. The screenshots are close-ups from the real GeoForge project rather than illustrations.

02.01

Know the target before clicking

The target is a project-contained acceptance run, not a publication-ready calibration study. The main scenario is a synthetic 50 ha rainfed corn field near Riesel. Public NASA POWER weather is used for 2010–2012, with a 20-year spin-up allowed by the KI. A successful run must leave real APEX outputs, validation metrics, plots, and provenance in the project folder.

- ✓ Model: APEX0806 through its documented Wine runtime on macOS.
- ✓ Inputs: weather, site, soil assumptions, crop management, and control files prepared by KI tools.
- ✓ Evidence: a completed real run, finite yields, a plausible water balance, plots, and provenance.

SUCCESS CHECK

You can restate the place, period, crop, area, and expected evidence in one sentence.

IF IT DOES NOT WORK

If your research question is still broad, begin with one site and a short period. Expand the case only after this first run succeeds.

02.02

Check that one AI connection is ready

- 1 Open AI Settings from the lower-left sidebar.
- 2 Choose either a ready local CLI (Claude Code, Codex, or Kimi) or a configured API such as DeepSeek.
- 3 For a local CLI, use Recheck after signing in through Terminal. For an API, save the key and run the connection check.

AI connections

Local agent CLIs

● **Claude Code · Ready**

signed in

● **OpenAI Codex · Ready**

signed in using ChatGPT

● **Kimi Code · Ready**

signed in

Recheck local CLIs

Default provider

(first usable) ▼

Anthropic API key

DeepSeek API key

...e560

OpenAI API key

OpenRouter API key

...d3a7

Save

Close

WHERE TO LOOK Read each provider's state here. Installed, signed in, and usable are separate checks.

SUCCESS CHECK

The provider is selectable in the chat header and no sign-in or update warning remains.

IF IT DOES NOT WORK

Open Terminal, run the provider's normal login command, return to GeoForge, and choose Recheck. Do not reinstall the scientific model to fix an AI login problem.

02.03

Create a project folder for this case

- 1 Choose New chat. GeoForge opens the New project chat window.
- 2 Use the suggested parent folder, or choose another writable local location.
- 3 Choose Create chat. GeoForge creates a new named subfolder; it does not modify existing files in the parent folder.

New project chat

Each chat has its own project folder for memory, inputs, model runs, outputs, and plots.

Create the project inside

/private/tmp/geoforge-doc-demo.rfQb06/projects

Choose...

GeoForge creates a new named folder inside this location. Existing files are not changed.

Create chat

Use default location

Cancel

WHERE TO LOOK The path is visible before the conversation begins. This folder becomes the durable memory and model workspace.

SUCCESS CHECK

A new conversation appears in the sidebar and the path shown at the bottom of the window points to its own folder.

IF IT DOES NOT WORK

Choose a folder under your user Documents directory. Avoid a read-only disk, the application bundle, or a cloud folder currently syncing thousands of files.

02.04

Describe the case in ordinary language

You do not need to list every APEX field. Give the agent the scientific scope and the evidence you want. The KI supplies the technical contract.

TYPE THIS REQUEST

I want to simulate a 50 ha rainfed corn field near Riesel, Texas, with APEX for 2010–2012. Use public weather data, run the real model, check water balance and yield, and save plots and provenance.

+ Files

◆ Skills

I want to simulate a 50 ha rainfed corn field near Riesel, Texas, with APEX for 2010–2012. Use public weather data, run the real model, check water balance and yield, and save plots and provenance.

Send

WHERE TO LOOK A concrete place, period, process, and output are enough to begin. The agent can ask for a real scientific decision later.

SUCCESS CHECK

The first agent response restates the scenario and says which KI it will use before running tools.

IF IT DOES NOT WORK

If the response is generic, add the missing place, dates, crop or process, and the result you expect. Do not paste dozens of unexplained model parameters.

02.05

Confirm the APEX KI

1 Choose the KI button in the chat header.

- 2 Search for APEX and select it. Leave Auto KI only when you genuinely want GeoForge to compare models.
- 3 Choose Apply. The selected KI is now fixed to this chat once the conversation starts running.

Knowledge Infrastructures for this chat

Leave empty for **Auto KI**. A green badge means the scientific software passed its check on this machine. You can still discuss KIs that need setup.

APEX

☐ ☒ **APEX** APEX (Agricultural Policy/Environmental eXtender) Manual setup

Apply

Use Auto KI

Close

WHERE TO LOOK The badge reports whether the software has passed its local check; it is different from whether this project's data are ready.

SUCCESS CHECK

APEX appears next to the KI mark in the chat header.

IF IT DOES NOT WORK

Open KI Library. If APEX is present but not verified, run its setup agent. If APEX is absent, import the KI package before returning to the chat.

02.06

Read the header before the run

- ✓ APEX tells you which KI is attached to this conversation.
- ✓ The verification badge refers to the scientific software on this Mac.
- ✓ Folder opens the current project's real directory in Finder.
- ✓ Project status opens the dynamic requirements, progress, data sources, calibration, and results panel.
- ✓ The right side shows the active provider and model for this chat.

APEX Riesel acceptance run

KI APEX

Manual setup

Folder

Needs you

AI CONNECTION

Local

API

Kimi Code

▼

kimi-for-coding

▼

WHERE TO LOOK These controls answer four different questions: which KI, whether software works, where files live, and what the project is doing.

SUCCESS CHECK

You can identify the selected KI, active AI provider, project folder button, and project-status button without reading the chat transcript.

IF IT DOES NOT WORK

If the provider or KI selectors are locked, the chat has already begun. Start a new chat to change the binding; do not silently switch providers halfway through one CLI memory session.

Open the dynamic project-data panel

Choose Project status. GeoForge reads the selected APEX KI and the current project, then groups only the requirements relevant to this run. It does not show one fixed generic checklist for every model.

Project status

One thing needs you 0 / 1 software verified

One data-source decision needs you

Goal
Run the real APEX0806 Riesel, Texas acceptance case for a 50 ha rainfed corn field using fresh NASA POWER weather for 2010-2012, a 20-year spin-up, the shipped KI tools, the real Wine-hosted binary, input and water-balance validation, plots, provenance, and a saved project calibration plan.

General KI: APEX0806

Project progress

- Understand** Understand the goal and choose a general KI
- Prepare** Check software and prepare suitable data
- Validate** Check the prepared model inputs
- Run** Run the real scientific model
- Results** Save and explain the results

Data for this run One thing needs you

Data categories are rebuilt from the active KI as the conversation changes.

Open all inputs

APEX0806

Built dynamically from APEX0806 KI - 22 requirements

Source data
Atmospheric CO2 concentration · Daily precipitation (PRCP) · Maximum air temperature (TMAX) · Minimum air temperature (TMIN) · +4 more

1 not ready yet

• Weather, observations, maps, and other raw material. Add your own data or let the agent help find a suitable source.
0 / 1 required data paths are ready.
3 local files in this group.
Inputs/forcing/cache · inputs · +2 more

Open folder

Resolve with agent

Spatial & field parameters

Continue in chat Add source data

WHERE TO LOOK The panel begins with the goal and five-stage progress, then shows current APEX data groups and their real readiness state.

SUCCESS CHECK

The goal mentions this Riesel APEX case, and the data section names APEX-specific requirements rather than a static demonstration list.

IF IT DOES NOT WORK

If the panel is empty, wait for the first agent response, confirm a KI, then close and reopen Project status. If it still fails, use the chat transcript's error and the project runs folder for diagnosis.

Inspect and resolve every input requirement

- 1 Find Source data. In this case it includes daily precipitation, minimum and maximum temperature, atmospheric CO₂, and other APEX weather inputs.
- 2 Open the category to see every requirement declared by the current KI. Open an individual item to inspect its purpose, KI, unit, expected file, format, source type, default, and valid range.
- 3 Choose Ask agent: what is this? for a read-only explanation tied to this project. The agent must distinguish item-level evidence from category-level readiness.

- 4 Choose Prepare/Find/Decide with agent for one item, or Resolve group with agent when the whole category should be handled together.
- 5 The agent first reuses and validates existing files, then follows the KI to download, convert, derive, or generate. Source URL, request parameters, checksums, units, and transformations stay in the project.

Source data

Atmospheric CO2 concentration · Daily precipitation (PRCP) · Maximum air temperature (TMAX) · Minimum air temperature (TMIN) · +4 more

Weather, observations, maps, and other raw material. Add your own data or let the agent help find a suitable source. 0 / 1 required data paths are ready. 8 requirements · open any item for details or ask the agent separately 3 local files in this group. inputs/forcing/cache · inputs · +2 more

1 not ready yet

Actions for this group

Open folder
Resolve group with agent

Atmospheric CO2 concentration

Drives radiation-use-efficiency CO2 response (bc1, bc2). ~420 ppm modern

Agent can handle

Purpose

Drives radiation-use-efficiency CO2 response (bc1, bc2). ~420 ppm modern

KI

APEX

Group

Source data

Unit

ppm by volume

Source type

forcing

Preparation

Agent can source, convert, or generate this

Ask agent: what is this?

Prepare with agent

Daily precipitation (PRCP)

May be read or stochastically generated (first-order Markov chain, Nicks 1974); partitioned to snowfall when mean of Tmax and surface soil temperature <= 0 C

Agent can handle

Maximum air temperature (TMAX)

Read or generated (Richardson 1981 multivariate model)

Agent can handle

Minimum air temperature (TMIN)

Read or generated

Agent can handle

Relative humidity (RHM)

Required for Penman PET; fraction not percent

Agent can handle

Solar radiation (SRAD)

Required for Penman-family PET. Expected ~10-25 MJ/m2/day; supplying W/m2 (multiply by 0.0864) silently distorts ET/water yield

Agent can handle

Continue in chat

Add source data

WHERE TO LOOK

Categories remain compact, but every KI requirement can be opened, explained by the agent, or resolved separately.

SUCCESS CHECK

Every declared item opens independently, both item buttons work, and Open folder leads to actual files under the current project.

IF IT DOES NOT WORK

Use Add source data for a private file. If a public download is blocked, wait for a structured Needs you request rather than copying an arbitrary replacement into the model directory.

02.09

Respond to a real "Needs you" decision

The example project detects a gap in a public station record. GeoForge does not bury the question in a long AI message. It opens one short decision card: fill the missing period from NASA POWER, or wait for

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your local station file.

?

Choose the weather data source

Needs you

The public station record has a gap in 2011. Choose how GeoForge should continue; no file will be replaced before you confirm.

☐

Use NASA POWER for the missing period
The agent downloads the missing dates, converts units, and records the source and checksums.

☐

I will add a local station file
GeoForge waits for your private file, then validates its fields and dates.

Optional note for the agent

Add context only if needed...

Continue with this choice

Ask about these choices

Not now

WHERE TO LOOK

Choose one option, ask the agent to compare them, or postpone. No file is replaced before you confirm.

SUCCESS CHECK

After you choose, the decision is written back into the conversation and the agent continues from the same project state.

IF IT DOES NOT WORK

Choose Ask about these choices. The agent should explain effects on provenance, consistency, and scientific interpretation in simple words, then present the choice again.

02.10

Add files or pin a skill without leaving the chat

- 1 Choose Files to open the system file picker, or drag files onto the message bar.
- 2 Choose Skills only when you want to pin a specific reusable procedure to this chat. Otherwise the agent may choose an appropriate skill automatically.
- 3 Attached files are copied into this project's input area and remain available to later messages in the same project.

+ Files

◆ Skills

Add a local observation file, or pin a plotting skill to this chat.

Send

WHERE TO LOOK

Files, skills, the prompt, and Send remain together at the bottom of the active chat.

SUCCESS CHECK

The file appears in the attachment list before sending, or the Skills button shows the number of pinned skills.

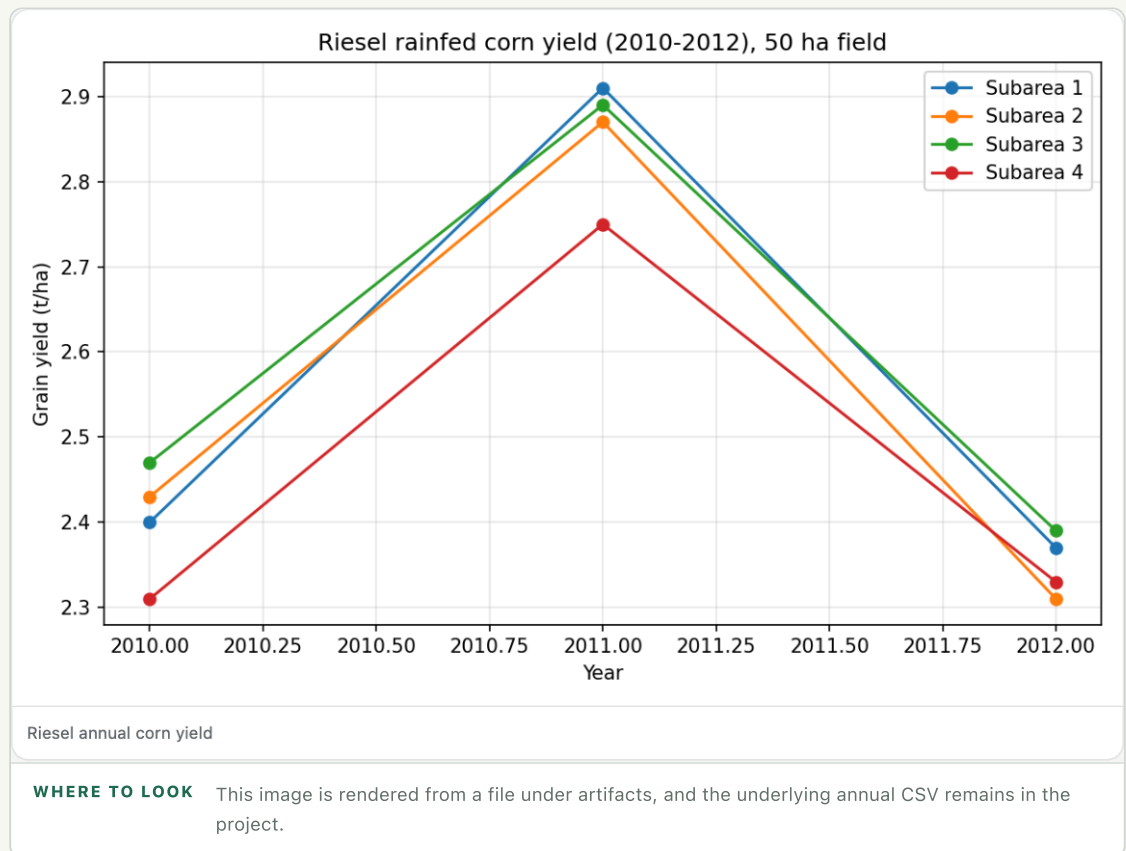
IF IT DOES NOT WORK

If a large file is rejected, copy it into the project inputs folder through Finder and tell the agent its relative path. Keep the original file outside the model's executable directory.

02.11

Verify the run from the saved yield plot

The completed acceptance case produced finite corn yields for 2010–2012. The actual-year range was 2.31–2.91 t/ha, with a mean of 2.54 t/ha. The Riesel reference template retains multiple subareas, so the saved plot contains four series even though the main configured subarea is 50 ha. This is a detail to inspect, not hide.



SUCCESS CHECK

The plot is visible in chat, opens as a saved artifact, and its values agree with the parsed CSV and the assistant's summary.

IF IT DOES NOT WORK

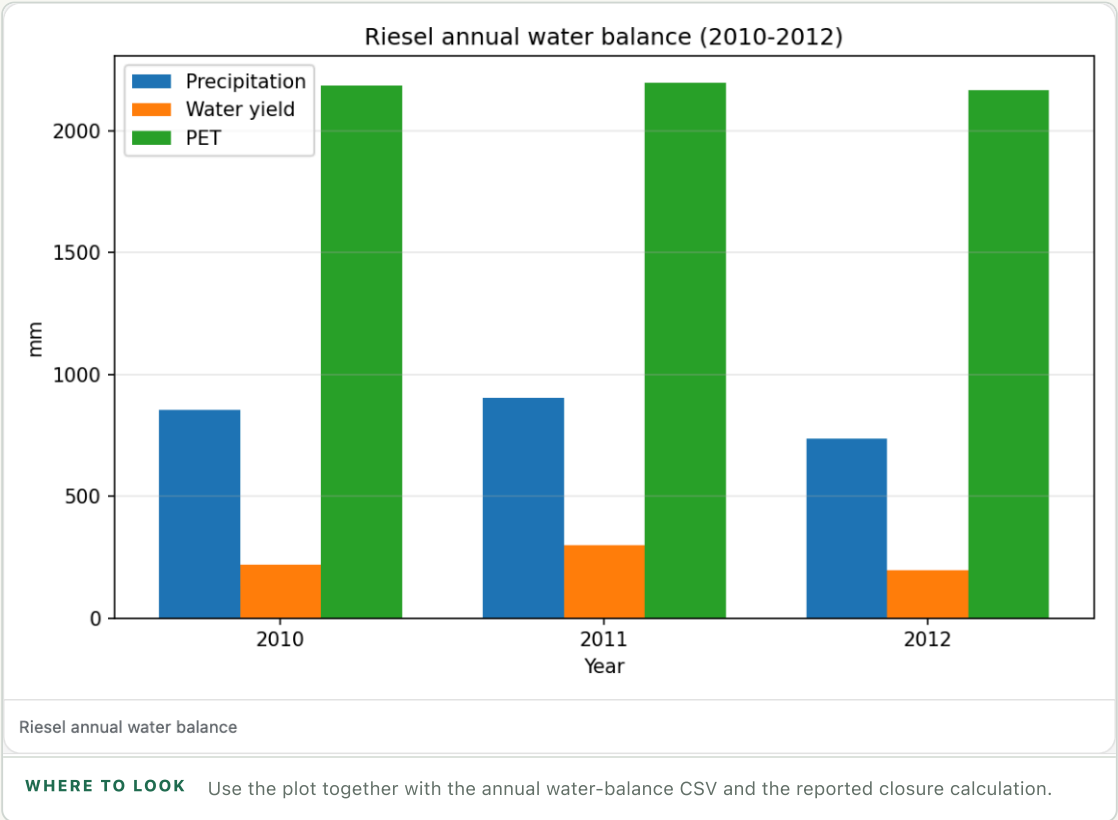
If a plot appears without a saved data table or model output, do not accept it as evidence. Ask the agent to identify the source file, parser, and exact artifact path.

02.12

Check the water balance, not only the crop plot

The case closed its water balance to 0.47% using precipitation against runoff, evapotranspiration, deep percolation, subsurface flow, and storage change. A small closure error supports internal consistency; it

does not by itself prove that every input or parameter is scientifically correct.



SUCCESS CHECK

The result reports the closure equation and value, all plotted values are finite, and the source CSV is saved.

IF IT DOES NOT WORK

Check units, date alignment, spin-up exclusion, missing-value handling, and whether storage change was included before changing model parameters.

Open the project folder and audit the evidence

- ① Choose Folder in the chat header. Finder opens this conversation's project directory.
- ② Open inputs to inspect source and prepared data; open outputs/APEX for the runnable model workspace and raw model outputs.
- ③ Open artifacts for parsed CSV files and plots; open runs for orchestration, progress, and execution records.
- ④ Open the provenance JSON and confirm the NASA POWER URL/query, checksums, transformations, executable, dates, and model version.

EXPECTED PROJECT STRUCTURE

project/	
├ inputs/	source and prepared inputs
├ outputs/APEX/	runnable APEX workspace and raw outputs
├ artifacts/	annual CSV files, yield plot, water-balance plot
├ runs/	orchestration and execution records
├ calibration/cases/	saved calibration plan
├ memory/	durable project memory
└ session.json	conversation and binding metadata

SUCCESS CHECK

Another researcher can identify the exact input, executable, command/workflow, raw output, parsed table, plot, and provenance record without relying on the chat prose.

IF IT DOES NOT WORK

If the chat says a file exists but Finder cannot find it, ask for the absolute project path and relative artifact path. Treat the file system as the source of truth.

Understand what the saved calibration plan means

This run saved a calibration case plan under calibration/cases with targets such as yield and water balance, candidate parameters, an optimization strategy, and a holdout design. The APEX adapter was still marked authoring-needed, so the calibration engine did not run. GeoForge correctly reported a plan, not a calibrated or transferable model.

- ✓ Plan saved: the workflow can be reviewed and completed later.
- ✓ Calibration run: requires a working project-specific adapter and real observations.
- ✓ Calibrated: requires successful model evaluations and objective results.
- ✓ Validated/promotable: additionally requires independent holdout validation to pass.

SUCCESS CHECK

The project contains a readable case plan, but the result does not claim calibration unless engine records and holdout metrics exist.

IF IT DOES NOT WORK

Add real observations, then use the calibration section in Project status to have the agent complete a project-local adapter. Never edit the shared engine or original KI to make one project pass.

NOTE

The honest stopping point is part of reproducibility. A saved plan is useful, but it is not a calibration result.

Install and launch on macOS

GeoForge is a local desktop application. The app bundle contains the interface and shared harness; individual scientific models may still require installation, licensing, or a separate download.

03.01

Install the app

- 1 Download the macOS release that matches your Mac. Apple Silicon is arm64; older Intel Macs use x86_64 where offered.
- 2 Unzip the release and move GeoForge Desktop.app to Applications.
- 3 On first launch, macOS may ask for confirmation. Use System Settings > Privacy & Security only if macOS identifies the downloaded app and you trust the release source.

03.02

Choose where work is stored

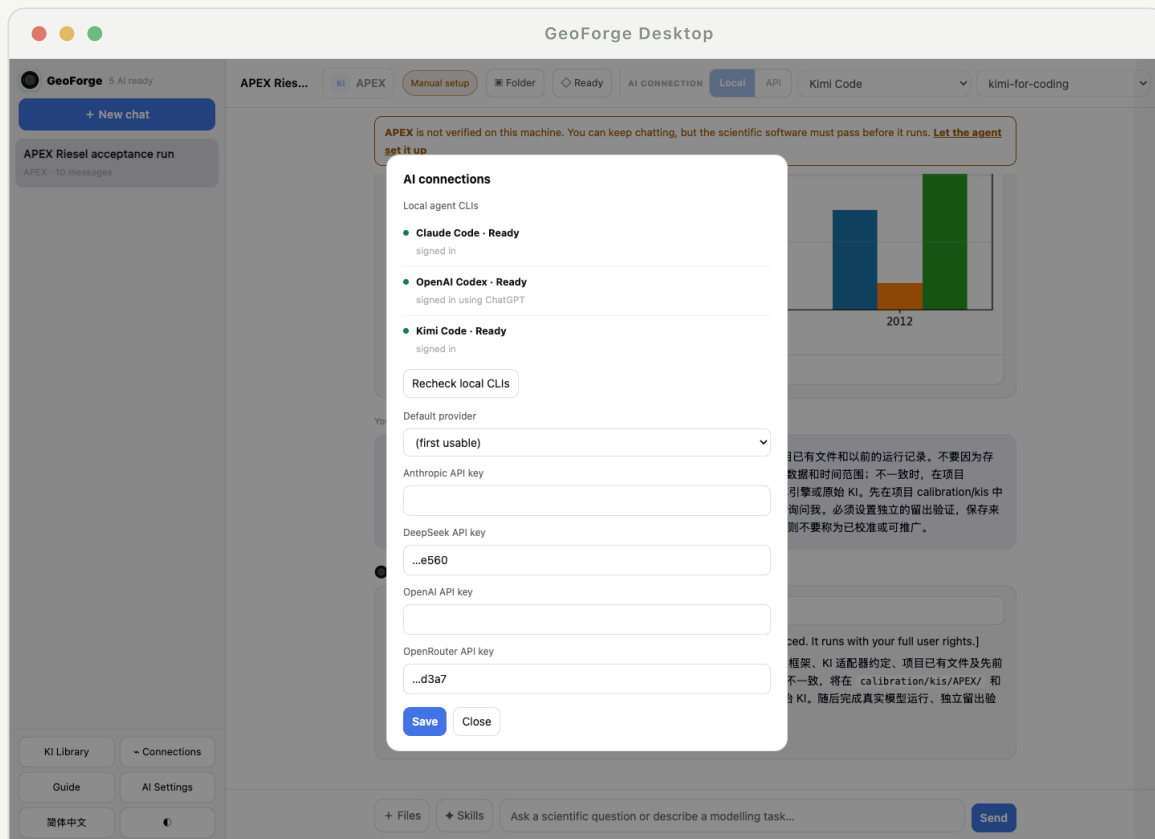
Every new conversation proposes a folder inside the GeoForge projects directory. You can accept it or choose another local folder. Keep projects outside the application bundle so updates never replace your work.

NOTE

For large models, choose a disk with enough free space. Avoid cloud-synced folders while a model is actively writing many files.

Connect an AI provider

GeoForge can use an authenticated local CLI or an API. The same KI harness and project tools sit underneath both; the provider supplies the reasoning and language interface.



PRODUCT SCREENSHOT Connection checks distinguish installed, authenticated, and usable.

04.01

Local CLI mode

- ✓ Claude Code: install and sign in in Terminal, then use Recheck in GeoForge.
- ✓ OpenAI Codex: install/update the CLI and complete its normal sign-in flow.
- ✓ Kimi: install Kimi Code, sign in, and select it from the local provider list.
- ✓ GeoForge starts the provider with the same user account but a controlled project working directory.

04.02

API mode

For DeepSeek or another supported API, enter the key in AI settings or expose it through the supported environment variable. A real one-message probe confirms that the key, endpoint, model, and network path work together.

IMPORTANT Installed does not mean signed in. Signed in does not mean the selected model is available. GeoForge reports these states separately.

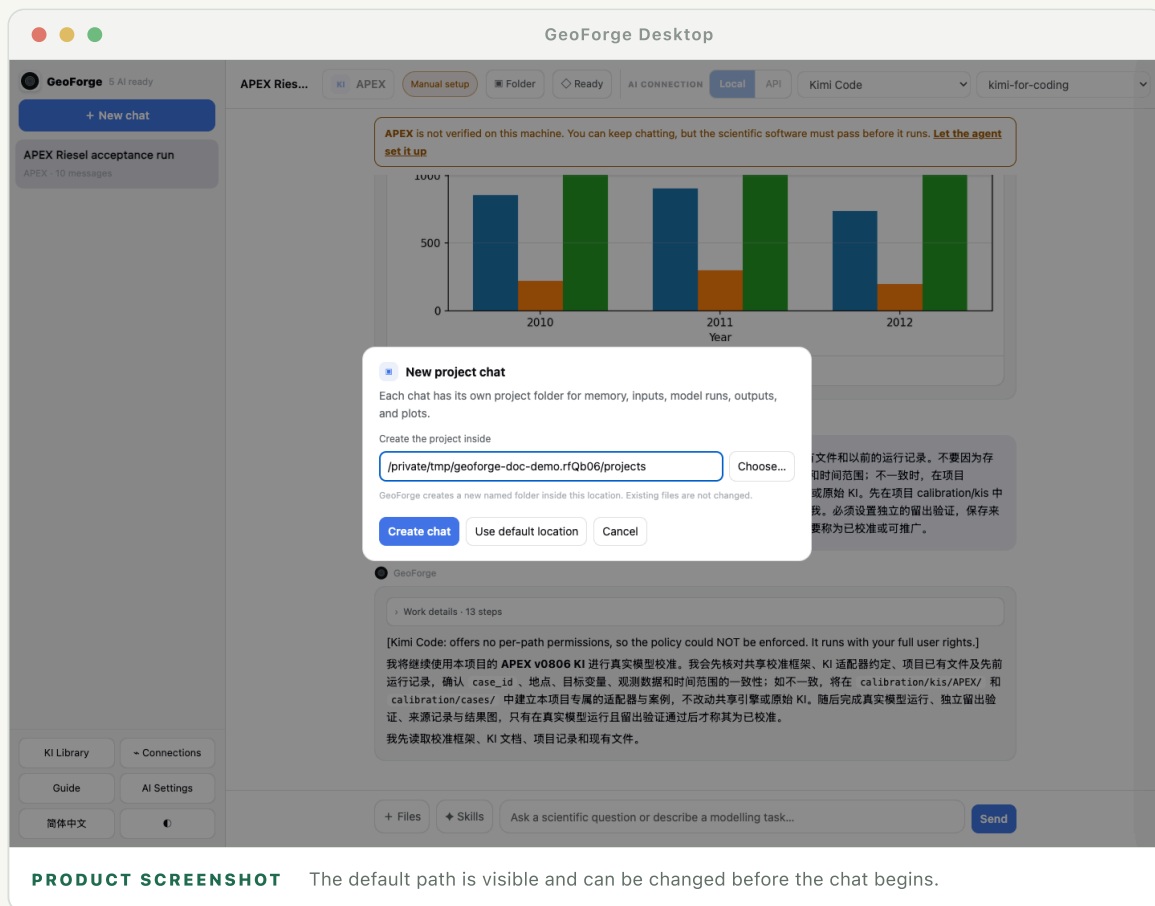
04.03

What is sent to a provider

The conversation, relevant KI instructions, and selected tool results may be sent to the active AI provider. Local model files are not automatically uploaded in full; the agent reads only what a tool exposes for the task. Review provider terms before using confidential data.

Projects, conversations, and memory

GeoForge does not depend on a provider remembering your project forever. Each conversation has a local project folder and a durable transcript. The provider's native session ID is kept when supported, but local project state remains the source of truth.



PRODUCT SCREENSHOT The default path is visible and can be changed before the chat begins.

What the project folder contains

- ✓ Conversation history and concise project state used to resume work.
- ✓ Selected KIs, execution policy, provider/session metadata, and progress events.
- ✓ Source files, prepared inputs, model workspaces, outputs, plots, and reports.
- ✓ Provenance: where data came from, how it was transformed, and which command produced an artifact.

Switching conversations while work continues

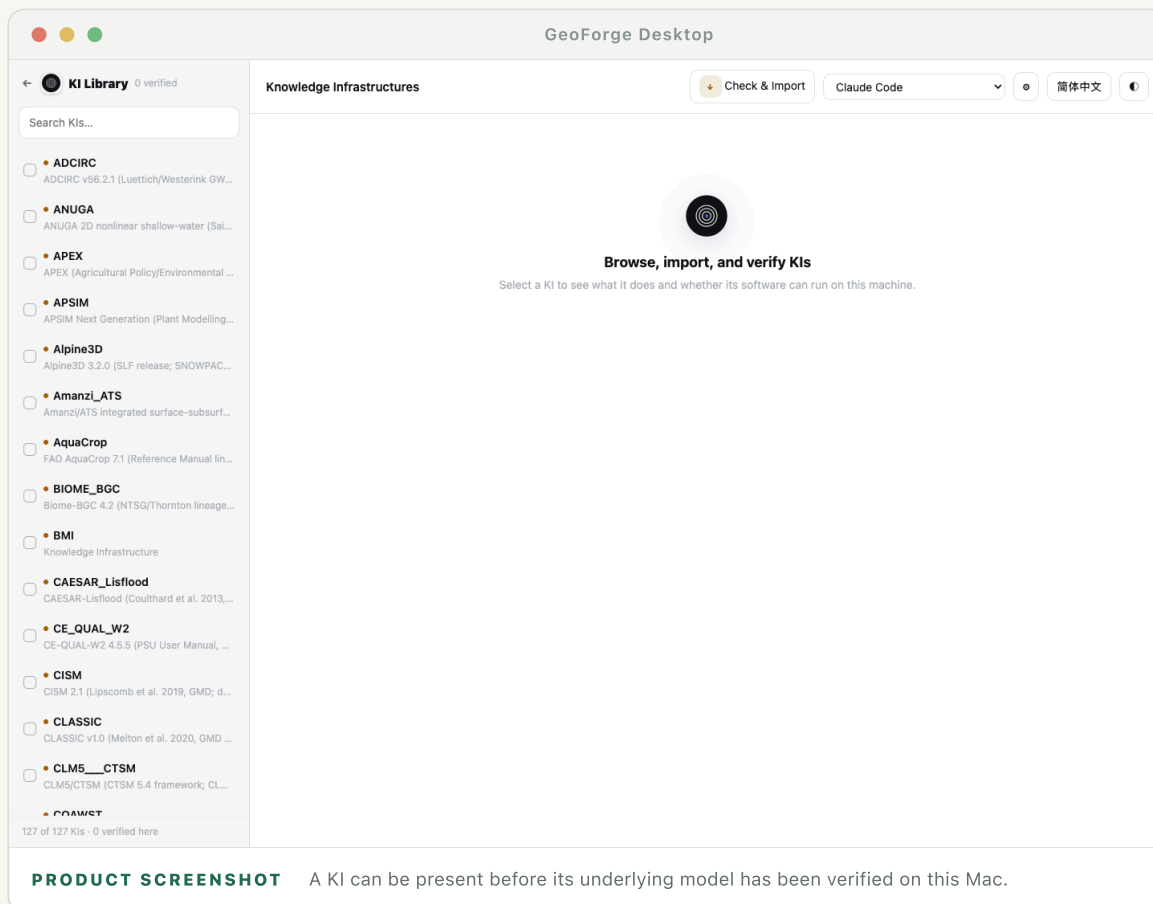
A running conversation should show visible activity and continue in its own task. You can create or open another conversation without treating the whole app as frozen. Returning to the original project shows its latest event, pending request, or result.

TIP

Use one conversation for one scientific objective. Start a new project when the question, study area, or model experiment changes materially.

Understand the KI Library

A Knowledge Infrastructure (KI) is the model-specific operating knowledge used by the agent. It can include setup recipes, tools, diagnostics, data contracts, examples, documentation summaries, and interpretation rules.



06.01

Three states that must not be confused

- ✓ KI available: GeoForge can read the model guidance and tools.
- ✓ Software verified on this Mac: the required executable passed the KI's real preflight check here.
- ✓ Project ready: this particular scenario has enough validated inputs and decisions to run.

Import and verify a KI

- ① Choose Import KI and select a KI package or repository folder.
- ② Run the verifier. It checks structure, declared tools, paths, and safety rules before import.
- ③ Review the scientific software, version, license, source, and last-check result.
- ④ Use Setup & verify when the KI exists but its model software is not yet ready.

NOTE

Verification is local and version-specific. A model verified on a Linux server is not automatically verified on macOS.

Work through chat

Users should not have to translate a scientific goal into dozens of file names. Start with the question. The agent uses the KI to discover technical requirements and surfaces only decisions that cannot be safely inferred.

07.01

A good first message

Include the process, place, period, scale, and desired comparison or output when you know them.

Example: "Simulate rainfed maize yield around Xinxiang for 2000–2020 and compare a warmer scenario with the baseline." It is fine to omit items you have not decided.

07.02

What the agent should handle

- ✓ Select or propose a suitable KI and explain the choice briefly.
- ✓ Inspect software readiness and repair ordinary setup problems inside the model workspace.
- ✓ Turn the KI data contract into a short project-specific data plan.
- ✓ Download public data, transform formats, generate derivable parameters, and validate inputs where permitted.
- ✓ Run the model, check outputs, make plots, and explain uncertainty without hiding failures.

07.03

When GeoForge should ask you

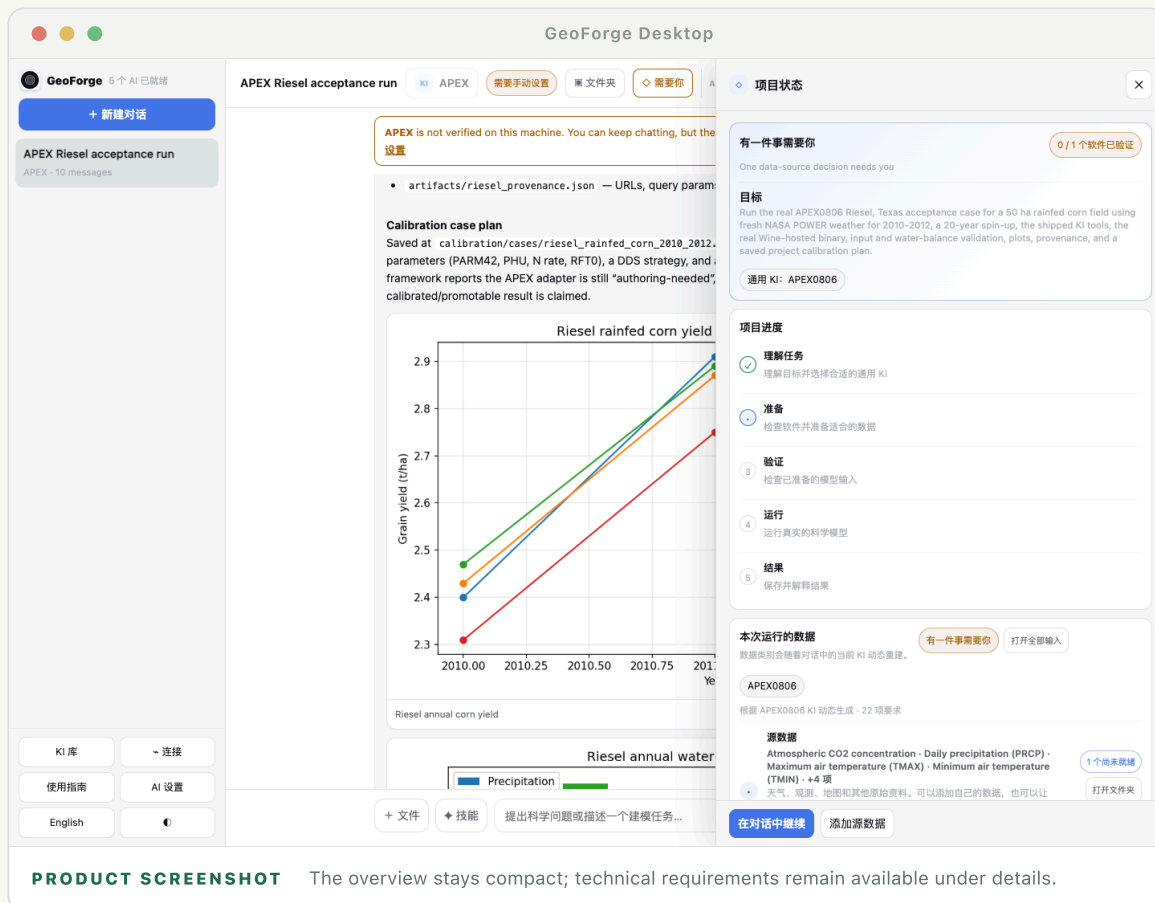
- ✓ A scientific choice has several defensible answers that change interpretation.
- ✓ A private file, login, license, protected download, or macOS permission is required.
- ✓ An action would write outside the project, install system software, or access a sensitive location.

TIP

You can always ask: "What did you assume, what is measured, and what is generated?"

Prepare project data

The data panel is generated after the agent understands the scenario and selects one or more KIs. It is not a static checklist and should not expose every internal model file as work for the user.



08.01

The four compact lanes

- ✓ Source data: weather, observations, maps, measurements, or other raw material.
- ✓ Spatial and grid parameters: catchments, layers, cells, HRUs, reaches, or point locations derived from source data.
- ✓ Model choices and calibration: parameters, objective functions, bounds, and defensible defaults.
- ✓ Initial and boundary conditions: the state at the start and the conditions around the modeled domain.

08.02

Use Resolve with agent

- ① Open a category to see its current state: ready, working, needs review, or needs you.
- ② Choose Resolve with agent. The agent reads the selected KI contract and the current scenario.
- ③ It finds permitted public sources, converts formats, derives technical files, and records provenance.
- ④ If a real choice or permission is needed, the task moves to Needs you with a focused request.

08.03

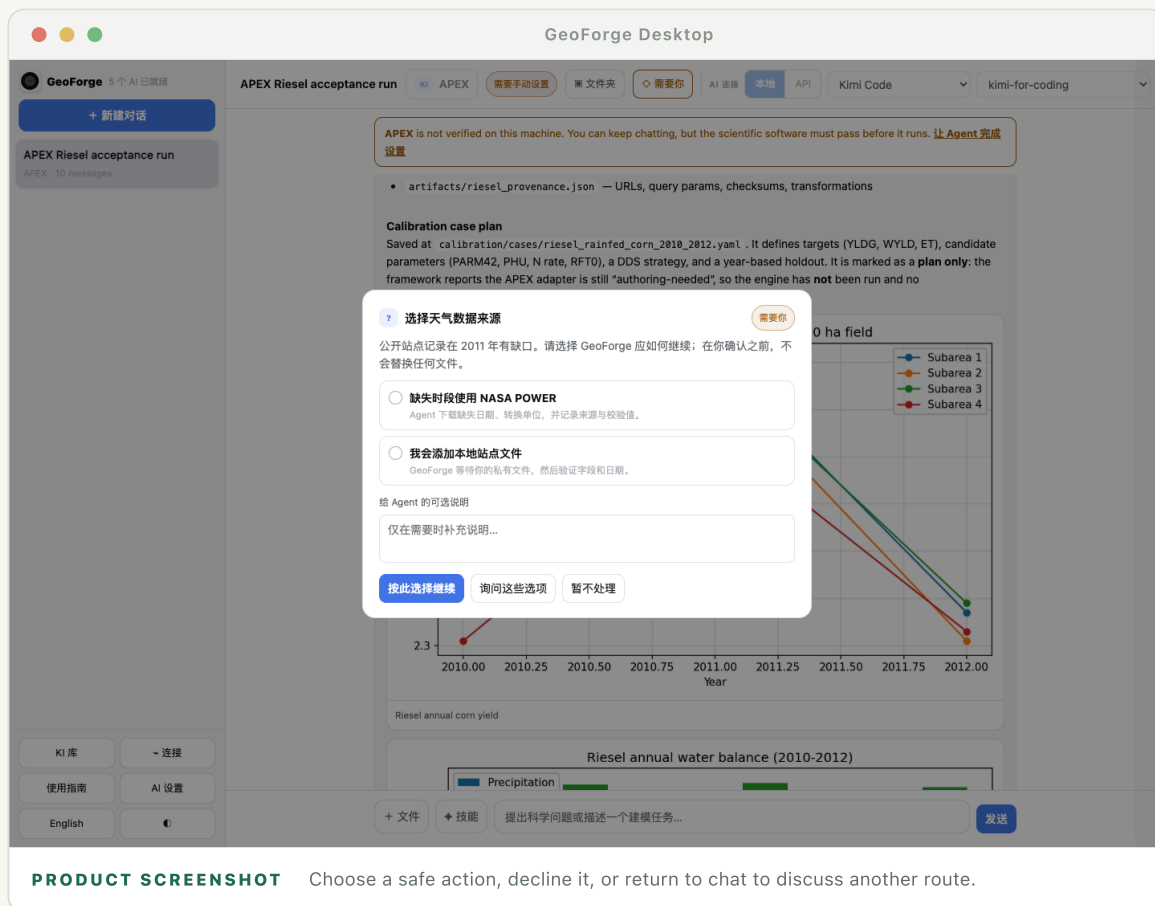
See the actual files

Use Open folder on the panel or the Folder button in chat. GeoForge opens the project location in Finder so you can inspect the source, prepared input, and output files directly. Detailed entries should show local path, source URL or citation, transformation, validation, date, and checksum when available.

IMPORTANT Ready means the requirement has a validated project path. It does not mean every value is observed or uncertainty-free.

Respond to “Needs you”

When the agent cannot safely continue alone, GeoForge should stop with a structured request—not bury a command inside a long chat message. The request explains what is blocked, why it matters, and the smallest action that unlocks progress.



PRODUCT SCREENSHOT Choose a safe action, decline it, or return to chat to discuss another route.

Common request types

- ✓ Permission: allow one file, folder, executable, or clearly scoped command.
- ✓ Download: open a publisher or data portal that requires terms, CAPTCHA, or an account.
- ✓ License: confirm that you have the right to install or run restricted software.
- ✓ Scientific decision: choose among alternatives whose consequences the agent explains.
- ✓ Private input: select a local file without making it public or silently substituting synthetic data.

Dynamic permission policy

GeoForge can grant narrowly scoped, recoverable actions during a project. A useful pattern is Allow once, Always allow for this project, Choose another file, and Deny. Broad access to the home directory or unrestricted shell execution should never be the default.

TIP

If an option is unclear, choose Discuss in chat. The agent should explain the risk and find a project-local alternative when possible.

Set up and verify model software

The setup agent follows the KI's KDT recipe in a model workspace. It materializes tools, prepares an environment, acquires or builds the correct version, and runs a real preflight. Ordinary failures should be diagnosed and retried automatically.

10.01

What verification proves

- ① The KI was materialized and its required shared tools are importable.
- ② Required runtime and system dependencies are present.
- ③ The intended software version was obtained or built for this platform.
- ④ The KI's real preflight starts the executable and checks expected behavior.

10.02

What the agent repairs itself

- ✓ A Git branch/tag mismatch, retryable clone failure, build directory, or project-local path.
- ✓ A missing project-local Python package such as `ki_tools_common`.
- ✓ A model binary that must be copied or linked into the controlled workspace.
- ✓ A preflight failure with a documented KI repair path.

10.03

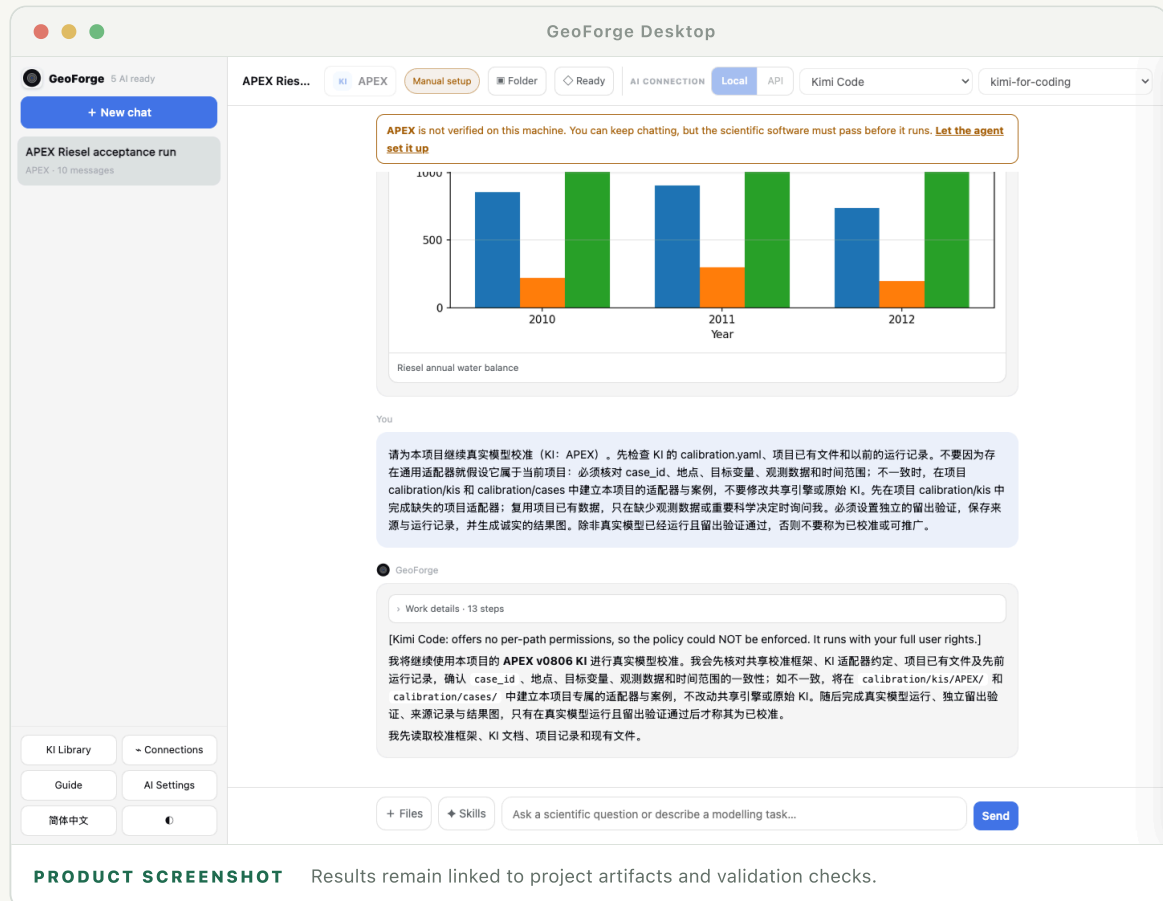
What still needs you

The agent must stop for licenses, protected downloads, credentials, macOS system permissions, or destructive/system-wide changes. Software verification remains separate from project data: missing weather or DEM data should not label a successfully installed model as broken.

IMPORTANT A setup log ending with "Load failed" is not enough. The UI should preserve the actual provider error, exit status, and last successful step.

Run, validate, and interpret

A successful GeoForge run has three layers: execution evidence, validation evidence, and scientific interpretation. The agent should make each layer visible and save it with the project.



Before execution

- ✓ Confirm study extent, dates, model version, units, and scenario assumptions.
- ✓ Validate required paths, file formats, missing values, ranges, coordinate reference systems, and temporal coverage.
- ✓ Freeze or record the prepared inputs so the run can be reproduced.

11.02

After execution

- ✓ Check the process exit status and model-specific completion markers.
 - ✓ Reject empty, stale, truncated, impossible, or non-finite outputs.
 - ✓ Run mass balance, benchmark, range, continuity, or reference-case checks declared by the KI.
 - ✓ Separate simulated results from observations, assumptions, and calibration targets.
-

11.03

Read the answer critically

A model result is conditional on the selected structure, data, parameterization, and scenario. Ask for sensitivity, holdout performance, missing processes, and the main reasons the result could change. GeoForge should report a failure or limitation directly rather than constructing a scientific story around an incomplete run.

Plots, tables, and project files

The agent can render plots and compact tables directly in chat. Every important visual should also be saved in the project so it can be reopened, regenerated, or used in a report.

12.01

Inline visualization

- ✓ Plots should state variables, units, dates, spatial scope, and whether values are observed or simulated.
- ✓ Calibration plots should distinguish calibration and validation periods.
- ✓ Maps should state projection, resolution, extent, and masking rules.
- ✓ A visual error should not erase the underlying model outputs; the agent can repair and re-render it.

12.02

Add your own files

Drag files onto the message composer or choose + Files to open the file picker. GeoForge copies or references them inside the project according to the selected policy, records their original names, and tells the agent where they are. Large folders should be added through the folder picker instead of pasted into chat.

TIP

Use Copy on a message for text, and Open folder for artifacts. Copying a displayed path is not the same as opening or granting access to it.

Build a reusable calibration workflow

Calibration is not a one-off conversation. GeoForge turns it into a saved workflow: a shared calibration engine, a model-specific KI adapter, and a project-specific case that can be rerun after changing bounds, observations, or budgets.

13.01

The three layers

- ✓ Calibration engine: algorithms, scheduling, checkpoints, metrics, and result storage shared across models.
- ✓ KI adapter: how parameters are written, how the model is launched, and how outputs become objective values.
- ✓ Project case: this study's parameters, bounds, observations, warm-up, calibration/validation periods, objective, and compute budget.

13.02

Create the workflow

- ① Ask to calibrate after a baseline run succeeds and the observed target is validated.
- ② Review proposed parameters, ranges, constraints, objective metrics, and holdout design.
- ③ Generate the adapter and case in the project's calibration folder.
- ④ Run a smoke test, then the selected budget. Failed candidates remain traceable rather than disappearing.
- ⑤ Save the best parameter set, full history, diagnostics, and a command or button for rerunning.

13.03

Update later

Open the saved calibration case, adjust allowed values such as iteration budget, parameter bounds, or target files, and choose Run calibration. GeoForge should create a new run record instead of overwriting the earlier result. Model or adapter version changes require a new validation check.

IMPORTANT Calibration can improve fit without improving truth. Always reserve independent data or a validation period and report both performances.

Skills and MCP connections

KIs explain how to operate scientific models. Skills add reusable methods such as plotting, statistical analysis, geospatial processing, or literature workflows. MCP connections give controlled access to external services such as GitHub.

14.01

Use skills in a conversation

Choose Skills beside the message box to review or pin skills for that conversation. You can also type a slash command or ask naturally, such as “make a publication-quality hydrograph.” The agent may automatically use a relevant installed skill, while the selected list makes your intended methods explicit.

14.02

Connect MCP services

- 1 Open Connections and choose an available MCP provider.
- 2 Review what resources and actions it exposes before authenticating.
- 3 Complete login in the provider's trusted flow and return to GeoForge.
- 4 Grant only the repositories, folders, or scopes needed for the project.

NOTE

A connection expands what the agent can access; it does not replace KI validation or project provenance.

Language, copy, and keyboard use

The interface language controls labels and guides. The conversation follows the language of your latest message, so a Chinese request should receive a Chinese answer unless you ask otherwise.

15.01

Language behavior

- ✓ Use English/中文 in the app to switch navigation, status, dialogs, and this guide.
 - ✓ Scientific names, model file names, commands, units, and paths remain unchanged when translating would make them incorrect.
 - ✓ Ask "answer in English" or "请全程使用中文" to override automatic conversation language.
-

15.02

Copy, paste, and shortcuts

- ✓ Select text normally and use Command-C / Command-V in the composer.
- ✓ Use Copy on an assistant message to copy its rendered text without internal tool markup.
- ✓ Press Command-K for a new task where shown; Enter sends and Shift-Enter inserts a new line.
- ✓ While an agent is running, navigation and New conversation remain available.

Privacy and reproducibility

GeoForge is local-first, but the selected AI provider and connected services may receive task context. Reproducibility also requires more than keeping outputs: the project must preserve the exact inputs and decisions that produced them.

16.01

Before using sensitive data

- ✓ Check the active provider, its data policy, and whether a local CLI still sends prompts to a cloud service.
- ✓ Remove secrets, personal identifiers, unpublished restricted data, and credentials from chat and ordinary project files.
- ✓ Use project-scoped permissions and disconnect external services when they are no longer required.
- ✓ Do not upload licensed papers or model binaries unless their terms permit it.

16.02

Minimum reproducibility record

- ✓ GeoForge, KI, model, adapter, and calibration-engine versions.
- ✓ Source data identifiers, URLs/citations, access dates, checksums, and licenses.
- ✓ All transformations, units, projections, missing-data decisions, and generated parameters.
- ✓ Execution commands, environment, random seeds, run dates, failures, and validation checks.
- ✓ Plots and tables linked to the exact result files they summarize.

TIP

Archive or export the whole project when a result supports a paper, report, or decision.

Troubleshooting and glossary

Start with the visible status and project event log. Do not repeatedly restart a task without reading the last real error: it may be waiting for authentication, permission, a download, or a scientific decision.

17.01

Common problems

- ✓ CLI says sign-in needed: run the provider's own status command in Terminal, confirm the same macOS user, then Recheck. A stale probe must not be treated as truth.
 - ✓ The app looks frozen: check the activity badge and project events. API calls and model runs should stream a stage, elapsed time, and Cancel/continue-in-background option.
 - ✓ No module named ki_tools_common: rematerialize shared tools into the project environment and verify the tool path before retrying.
 - ✓ Preflight failed because data is missing: distinguish software preflight from project-input validation. Verify software first, then prepare scenario data.
 - ✓ Git clone failed: the setup agent should inspect available tags/branches, network status, and partial directories, then retry safely. Ask you only for authentication or protected access.
 - ✓ Copy/paste does not work: click inside the composer, use the Edit menu or Command-C/Command-V, and report the exact field if selection remains blocked.
-

17.02

Glossary

- ✓ KI — Knowledge Infrastructure: model-specific knowledge, tools, checks, and data contracts used by the agent.
 - ✓ KDT — the executable setup/diagnostic recipe that makes a KI's software usable on a machine.
 - ✓ Preflight — a real check performed before a full project run; it should prove the intended software can execute.
 - ✓ Project run — one traceable execution with fixed inputs, configuration, outputs, and validation.
 - ✓ Adapter — the model-specific bridge between a shared framework and one model's files, command, and outputs.
 - ✓ Holdout — observations or a period excluded from calibration and used for independent evaluation.
 - ✓ Provenance — the recorded origin and transformation history of data, parameters, code, and results.
-

17.03

Report a useful bug

Include the GeoForge version, macOS version and chip, active provider/model, KI, project stage, exact visible error, and the relevant project event/log. Remove API keys and private data. A screenshot is useful, but the underlying text and timestamp are usually more diagnostic.



Turn scientific knowledge into runnable,
inspectable, reproducible workflows.

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